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Supporting Information for

Circulating cell-free RNA signatures for the characterization and diagnosis of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS)

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This PDF file includes:

Figures S1 to S5
Legend for Dataset S1

Other supporting materials for this manuscript include the following:

Dataset S1

Supplemental Figures

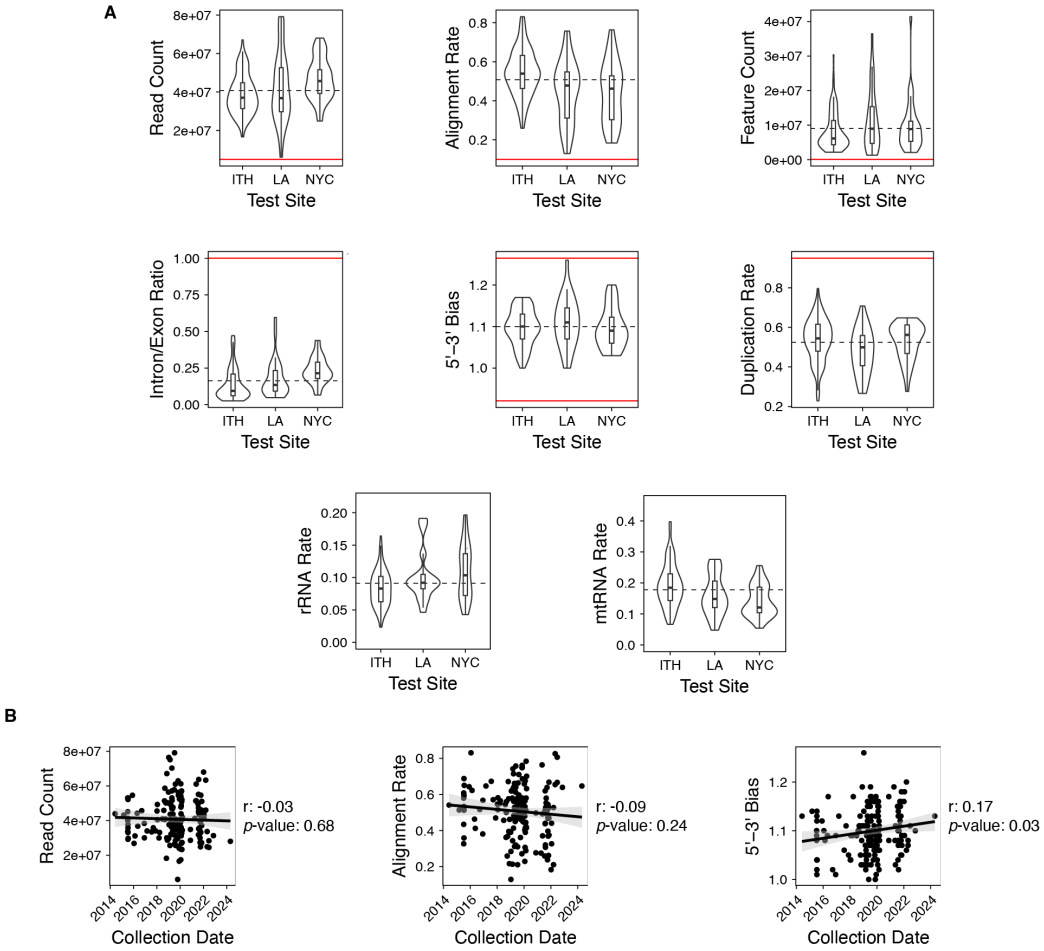


Fig. S1. Quality control metrics of cfRNA sequencing. **A.** Samples which passed the quality filter all had greater than 5 million total reads, greater than 100,000 total feature counts, greater than 10% alignment rate, less than 1-fold intron to exon ratio to estimate DNA contamination, less than 3 standard deviations from the mean of the 5'-3' bias to estimate RNA degradation, and less than 95% duplication rate. rRNA rate and mtRNA rates between collection/ test sites are also shown. **B.** Correlation of plasma sample collection date and cfRNA sequencing quality control metrics including read count, alignment rate, and 5'-3' bias.

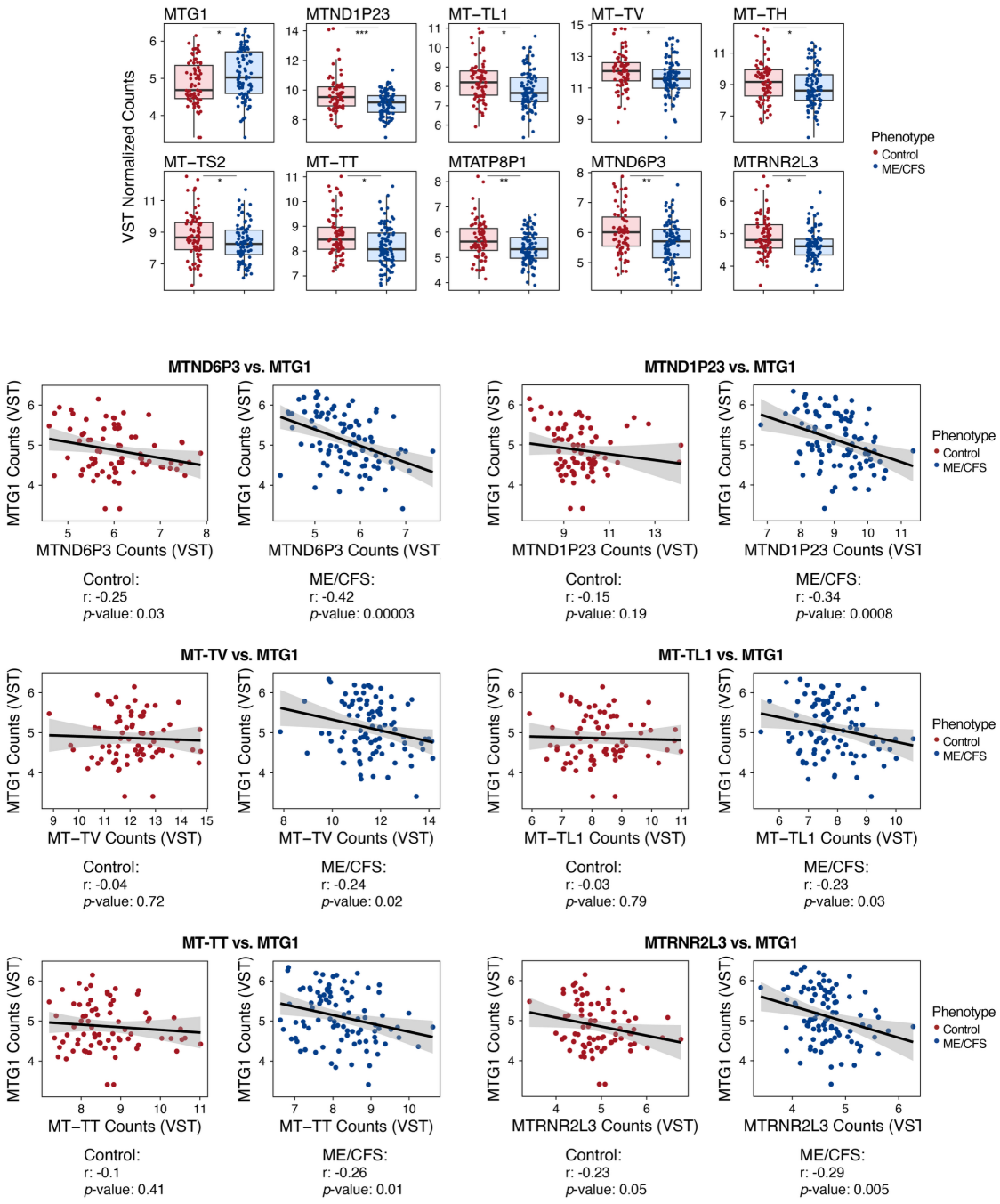


Fig. S2. Expression and correlations of mtRNA transcripts and MTG1. Variance stabilizing transformation (VST) gene expression counts of MTG1, mitochondrial ribosome associated GTPase 1, and 9 mtRNA genes that are significantly less abundant in cases. Asterisks indicate statistical significance by Mann-Whitney U test, using p-values as follows; * : $p < 0.05$, ** : $p < 0.01$, *** : $p < 0.001$. Correlation plots show varying trends of expression between controls and cases for MTG1 and mtRNA genes. Cases show a significant inverse correlation in MTG1 expression and the selected mtRNA genes whereas controls do not.

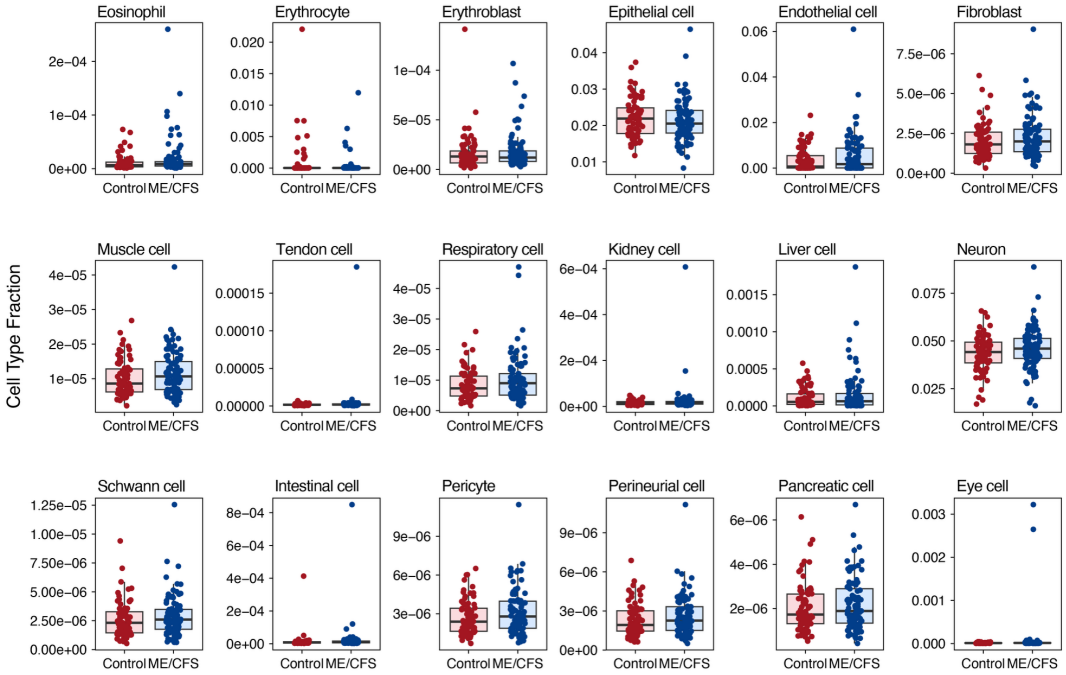


Fig. S3. Solid tissue derived cfRNA fractions. The Tabula Sapiens atlas was used as a single-cell RNA-seq reference input into BayesPrism to deconvolve the solid tissue derived fraction of cfRNA in cases and controls. There are no significant differences in solid tissue cell types between phenotypes, and the total fraction of solid tissue cell types.

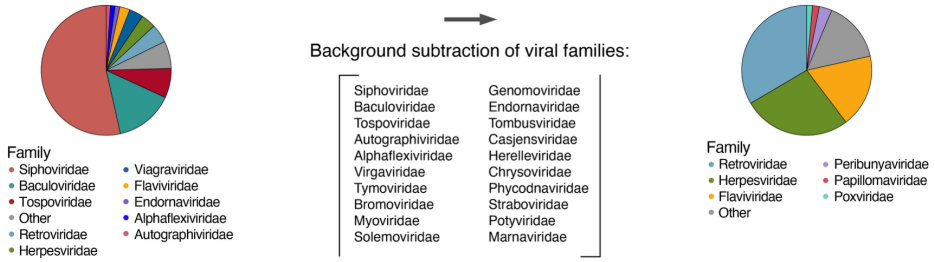


Fig. S4. Circulating viral RNA background removal. Left pie chart includes all viral families prior to filtering. 20 non-mammalian infecting viral families were bioinformatically removed from both cases and controls. Right pie chart includes the remaining viral families after background subtraction.

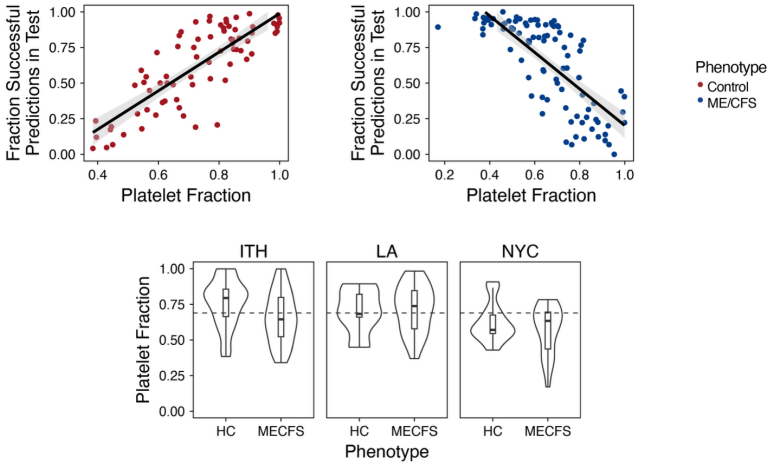


Fig. S5. Platelet fraction and classification success. The fraction of platelet derived cfRNA, as determined by BayesPrism deconvolution, in each sample correlates with the fraction of successful predictions for each sample when placed in the test set. Controls have a positive correlation of these metrics (Pearson r = 0.85, p-value < 1e-23), whereas cases have an inverse correlation (Pearson r = -0.82, p-value < 1e-20), indicating a phenotypic difference in platelets as determined by machine learning. Violin plot of platelet fraction by test site and phenotype shows this correlation is not dependent on collection/ test site.

Dataset S1 (separate file). Subject demographics for participant data. Duration of ME/CFS, BMI, and Age are shown as ranges placed into bins. Age Bins: Bin 1: >18 to <=35; Bin 2: >35 to <=45; Bin 3: >45 to <=55; Bin 4: >55 to <=70. BMI Bins: Bin 1: >0 to <=18.5; Bin 2: >18.5 to <=25; Bin 3: >25 to <=27; Bin 4: >27 to <=30; Bin 5: >30. VO2 peak Bins: Bin 1: >0 to <=15; Bin 2: >15 to <=20; Bin 3: >20 to <=25; Bin 4: >25 to <=30; Bin 5: >30. GH Bins: Bin 1: >0 to <=20; Bin 2: >20 to <=40; Bin 3: >40 to <=60; Bin 4: >60 to <=80; Bin 5: >80. MFI20_Total Bins: Bin 1: >0 to <=20; Bin 2: >20 to <=40; Bin 3: >40 to <=60; Bin 4: >60 to <=80; Bin 5: >80. Duration of ME/CFS Bins: Bin 1: >0 to <=5; Bin 2: >5 to <=100.